

Bequest specification for a housing–fertility lifecycle model: what the motive should depend on

Fertility–Housing project

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1 The question

The object at issue is a primitive: the utility a household derives from the estate it leaves at death, as a function of the estate size and of the household’s completed number of children n . The active model gives this terminal payoff the form

$$v(b, n) = \theta_0 \max\{1 + \theta_n n, 0\} \left[\frac{(\theta_1 + b)^{1-\sigma}}{1-\sigma} - \frac{\theta_1^{1-\sigma}}{1-\sigma} \right], \quad \theta_1 = 0.01, \sigma = 2, \quad (1)$$

where b is the gross estate realized at certain death at age 82 and every household is endowed with the motive.¹ Two features of (1) do economic work: the multiplier $\max\{1 + \theta_n n, 0\}$ makes the *strength* of the motive rise linearly in the number of children, and the near-zero shifter θ_1 makes the term a nearly homothetic warm glow rather than a luxury good that only the rich act on.

The design question has an extensive and a functional part. Extensively, is the motive a property of parents, so that childless households leave estates only accidentally? Functionally, if the motive is present, does its strength rise with the number of children, and is a linear estate multiplier the right way to encode that? The two parts have different answers, and neither answer is the current specification.

2 Evidence

(a) The extensive margin of the motive. The best direct evidence finds at most an extensive shifter, not a strong gradient in n . Kopczuk and Lupton (2007) estimate that about three-quarters of elderly singles hold a bequest motive; children raise the probability of a motive from 63% to 79%, significant only at the 10% level, and the remaining child covariates (income, education, proximity) are mostly insignificant. They read the evidence as an egoistic motive but explicitly reject the older assumption, due to Hurd (1989), that only parents have a motive at all—an assumption Hurd used to rationalize faster parental decumulation as accidental bequests. The margin the data support is that parents are more likely to *have* a motive, not that the motive scales with how many children they have.

(b) Children matter, but through transfers and division. The operative child channel in the modern evidence is inter vivos giving, not the shape of the terminal estate utility. Kværner (2023) uses quasi-random parental cancer diagnoses in Norwegian administrative data: a diagnosis raises children’s wealth by roughly 10% while dynasty wealth is unchanged, which identifies deliberate, child-directed *transfers* and re-reads Hurd’s faster decumulation as giving rather than indifference. Against a strong altruistic reading, Altonji et al. (1997) find that redistributing one dollar from parent to child moves the transfer by less than thirteen cents, rejecting the sharp Barro–Becker benchmark. McGarry (1999) shows that inter vivos transfers are compensatory—they flow to lower-income children—yet bequests are divided *equally* in 83% of wills, and Wilhelm (1996) finds 88% equal division in estate records, regardless of child income. The terminal estate is therefore close to a mechanical equal split; the child-contingent action is in the timing and targeting of transfers before death.

¹Form transcribed from `code/model/intergen_housing_fertility/solver.py` (`bequest.utility_vec`); not modified here.

(c) **Accumulation by asset class.** Whether parents “save more” depends entirely on the asset, and the pattern favors housing as the child-directed stock. Scholz and Seshadri (working paper) find that children have a large negative effect on financial net worth over working life: families with children consume more and optimally hold less retirement wealth, a crowd-out rather than an absence of caring. Boar (2021) shows the mirror image in consumption: parental consumption responds to the *child’s* permanent income risk with an elasticity of -0.06 to -0.08 , precautionary saving held as insurance for the child. Bernheim (1991) infers bequest motives from the life-insurance and annuity choices of the elderly, a further child-contingent protective margin. Housing runs the other way: the housing budget share rises with the number of children (Coeurdacier et al., 2023), consistent with this repository’s own PSID birth event studies. The synthesis is that “parents save more” is true in houses, insurance, transfers, and buffers and false in measured financial wealth—and housing is exactly the child-directed asset the model already carries.

(d) **Structural conventions.** The structural literature encodes the motive as a child-blind luxury warm glow. De Nardi (2004) introduces the luxury shifter that makes bequests a good the rich act on; Lockwood (2018) and Ameriks et al. (2011) estimate the same $v(X) = \frac{a}{1-\sigma}(X/a + \phi)^{1-\sigma}$ form, with ϕ the threshold below which nothing is left. De Nardi et al. (2025) find bequests behave as luxuries concentrated among rich couples, with children not even a conditioning variable in the saving heterogeneity. Closest to this project, Nakajima and Telyukova (2017) estimate a child-blind warm glow inside a housing model, $\gamma(a + \zeta)^{1-\sigma}/(1-\sigma)$, with strength $\gamma = 0.43$ and shifter $\zeta \approx \$19,600$ —a luxury threshold of order a quarter of annual household income. Even Kværner (2023), the paper with the strongest child-directed evidence, models the motive itself child-blind and puts children in the transfer block.²

3 Model evidence

The current block fails for a reason internal to the model, and the failure is sharp. With certain death at age 82, no survival risk, and no downsizing force, the cheapest way to hold an estate is housing equity, so the motive resolves into a single instruction: die owning the largest possible house, leveraged. Table 1 shows what switching pure warm glow on does at the clean frontier coordinates.

object	$\theta_0 = 0$	$\theta_0 = 0.3$	data target
ownership at ages 74+	0.606	0.981	~ 0.76
mean liquid wealth 74+ / 62–74	+0.17	−0.33	—
old nonhousing wealth median	2.33	3.30	2.23
TFR	2.004	2.000	1.918

Table 1: Fixed- θ profiles, $\theta_n = 0$, tight evaluator. Warm glow drives old-age ownership to 98% and mean liquid wealth at 74+ negative: households borrow against the house, consume, and bequeath the house.

The old-age ownership target (0.764, weight 160) makes this arithmetically unaffordable, and it is right to bind: the data say old households do not all keep their houses. Reoptimizing everything else with $\theta_0 = 0.3$ forced lands at 9.84–10.42 against a verified no-bequest frontier of 6.96 tight, so the reoptimized price of pure warm-glow bequests is about +2.9 loss points; even the smallest tested warm glow, $\theta_0 = 0.05$, costs +2.0. The linear child multiplier is a second, separable catastrophe, multiplying the penalty roughly sevenfold on the audit’s accounting: at $\theta_0 = 0.3$ the loss rises from 11.43 under pure warm glow to 96.83 with the multiplier on, as the estate becomes a fertility subsidy stacked on the housing-retention channel.³ Inverse-variance reweighting does not reverse the verdict: under SE-proportional weights the no-bequest baseline scores 345.1

²Verification status, following the round-3 source memo’s tags. Independently verified today [V]: the negative sign of children on financial net worth (Scholz and Seshadri), the -0.06 to -0.08 child-risk consumption elasticity (Boar), the rising housing budget share in n (Coeurdacier et al.), and the life-insurance reading (Bernheim, at search level). Extracted from paper pages or abstracts today but not yet re-checked against the papers [E], and load-bearing: Kopczuk and Lupton’s 63% $\rightarrow$ 79% and 10% significance, Nakajima and Telyukova’s $\gamma = 0.43$ and $\zeta \approx \$19,600$, Kværner’s $\sim 10\%$ wealth effect, and McGarry’s 83% / Wilhelm’s 88% equal-division shares. Spot-check these before paper use.

³Diagnostic value $\theta_n = 0.375$; the production default is $\theta_n = 0.25$. The parameter is an exact null direction at $\theta_0 \approx 0$, since the multiplier scales a term whose level is zero.

against 407.2–447.8 for the best bequest chain, because precision weighting loads even harder on the ownership moments that bequests wreck. Bequests are not rejected by the data or the targets; they are rejected by an accounting identity—estates can only be houses—that the model change in Section 4 is designed to break.

4 Proposed specification

Baseline. Replace (1) with a child-blind luxury warm glow that parents, and only parents, carry:

$$v(b, n) = \mathbf{1}\{n \geq 1\} \theta_0 \frac{(\theta_1 + b)^{1-\sigma}}{1-\sigma}, \quad \theta_1 \text{ estimated.} \quad (2)$$

The indicator encodes the one margin the data support—parents are more likely to have a motive (Kopczuk and Lupton, 2007; Hurd, 1989)—while the strength is child-blind in line with De Nardi (2004); De Nardi et al. (2025); Nakajima and Telyukova (2017), and θ_1 is promoted from 0.01 to an estimated luxury scale of order a quarter of annual income, anchored near the Nakajima and Telyukova (2017) shifter. Identification is clean: with the two free bequest parameters (θ_0, θ_1) , the old nonhousing wealth median (target 2.23) disciplines the luxury threshold and the decumulation moment below disciplines the strength.

Where the child channel goes. The intensive dependence on n moves out of the estate utility and into inter vivos down-payment transfers at family formation, where the causal and descriptive evidence places it (Kvæerner, 2023; McGarry, 1999), and where the down-payment-gift literature locates it for constrained first-time buyers.⁴ This is also where a housing–fertility mechanism naturally lives, since the transfer relaxes the very $(1 - \phi)$ down-payment constraint that couples young ownership to childbearing. Dropping θ_n costs no identification: it is a null direction at $\theta_0 \approx 0$, no literature supports the linear estate multiplier, and its economic content is reassigned to the transfer block, which carries its own identifying moment in birth-timed gift incidence.

Equal-division robustness variant. If a child term in the estate is wanted at all, the disciplined form comes from equal division rather than a free slope. Under equal division each of n children receives b/n ; aggregating n identical CRRA continuation values with a Barro–Becker altruism weight n^ϵ (Barro and Becker, 1989) gives

$$\theta_0 n^\epsilon \frac{(\theta_1 + b/n)^{1-\sigma}}{1-\sigma} \xrightarrow{\theta_1 \rightarrow 0} \theta_0 n^{\epsilon+\sigma-1} \frac{b^{1-\sigma}}{1-\sigma}, \quad (3)$$

so equal division plus altruism collapses to a warm glow whose child-count scale is $n^{\epsilon+\sigma-1}$, not a free linear multiplier. For $\sigma = 2$ the exponent is $\epsilon + 1$, which lies in $[1, 2]$ for $\epsilon \in [0, 1]$ —a bounded, micro-founded range in place of the unrestricted θ_n ; the exact form $\theta_0 n u(\theta_1 + b/n)$ is the $\epsilon = 1$ endpoint. This variant is justified by the equal-division fact (McGarry, 1999; Wilhelm, 1996), not by a stronger motive for parents, and it should be reported as robustness, never as the baseline.

A first bounded diagnostic (14 July). A 53-solve exercise tested the parent-gated form (2) at $(\theta_0, \theta_1) \in \{0.05, 0.30\} \times \{0.25, 1.00\}$, with and without an owner loan-to-value cap tapering linearly from age 66 to zero at 82, holding the live 15-moment objective and tight evaluator fixed.⁵ The gated form alone is loss-neutral: the best positive-bequest point scores 6.9774 against the no-bequest frontier 6.9647, so the respecified utility neither helps nor harms the fit while the exit margin is missing. The taper arm reaches 4.3718, but the gain is an artifact of where the data are measured: the forced schedule generates an ownership cliff—0.860 at age 70, 0.561 at 74, near zero at 78—that matches the 65–75 ownership average (0.766 against 0.764) precisely because the collapse lands just outside the measured window, while the 74+ / 62–74 liquid-wealth ratio stays at 1.016, so the model still produces no retirement decumulation. Households are pushed out of houses without being induced to run down wealth, the opposite of the missing margin. The diagnostic

⁴Engelhardt and Mayer, down-payment-gift evidence; the exact year and outlet are not verified from within the repository and the citation is deliberately left unpinned.

⁵Packet: `output/model/intergen_bequest_calibration_exercise_20260714/`; summary in the top section of `CALIBRATION_STATUS.md`.

therefore sharpens the prerequisite: the exit margin must be an empirically pinned mortgage-balance or downsizing profile, not a forced linear-to-zero loan-to-value schedule, and the moment set must include 75+ ownership and a 75 + / 62–74 decumulation ratio before θ_0 is re-estimated.

Structural prerequisites (model changes awaiting approval). None of (2)–(3) should be estimated until the accounting identity of Section 3 is broken. That requires an old-age exit margin—an externally pinned post-retirement amortization schedule or survival risk before age 82—so that estates can be financial rather than only housing, and a 75 + / 62–74 decumulation moment so that θ_0 is identified against β rather than confounded with it. To be explicit about scope, the following are proposals, not implemented changes: (i) the old-age exit margin; (ii) the new decumulation target; (iii) the $\mathbf{1}\{n \geq 1\}$ respecification of the motive; (iv) promoting θ_1 to an estimated parameter; and (v) removing the $(1 + \theta_n n)$ multiplier. Items (i) and (ii) are the load-bearing model changes and should be approved before θ_0 is re-estimated.

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